



Fig. 4: Receiver section of the wireless gesture-controlled robot

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - ATmega328 microcontroller
- IC3 - LM1117-3.3, 3.3V regulator
- IC4 - HT12E, 2nd series encoder
- IC5 - HT12D, 2nd series decoder
- IC6 - L293D, dual H-bridge motor driver

LED1-LED4

- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-mega-ohm
- R2 - 10-kilo-ohm
- R3 - 750-kilo-ohm
- R4-R7 - 220-ohm
- R8 - 47-kilo-ohm

Capacitors:

- C1, C2 - 22pF ceramic disk
- C3 - 0.1 μ F ceramic disk
- C4 - 0.33 μ F ceramic disk
- C5 - 10 μ F, 16V electrolytic

Miscellaneous:

- CON1, CON3 - 2-pin connector
- CON2 - 6-pin connector
- X_{crystal}1 - 16MHz crystal
- TX1 - 433MHz transmitter module
- RX1 - 433MHz receiver module
- M1, M2 - DC-gear motor, 100rpm
- S1, S2 - On/off switch
- Batt.1 - 9V PP3 battery
- Batt.2 - 4.5V, 1.5Ah lead-acid battery
- ANT.1, ANT.2 - 17cm long single-strand wire antenna
- ADXL335 - 3-axis accelerometer

provide resistance against acceleration forces. Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and plates attached to the moving mass.

Fixed plates are driven by 180° out-of-phase square waves. Acceleration deflects the moving mass and unbalances the differential capacitor, resulting in a sensor output whose

TABLE I
Movement of Robot and Decoder Outputs

Robot (accelerometer)	Input 1 (D11)	Input 2 (D10)	Input 3 (D9)	Input 4 (D8)
Forward (-Y)	0	1	0	1
Backward (+Y)	1	0	1	0
Right (+X)	1	0	0	1
Left (-X)	0	1	1	0

amplitude is proportional to acceleration. Phase-sensitive demodulation techniques are then used to determine the magnitude and direction of the acceleration.

L293D. This is a 16-pin DIP package motor driver IC (IC6) having four input pins and four output pins. All four input pins are connected to output pins of the decoder IC (IC5) and the four output pins are connected to DC motors of the robot. Enable pins are used to enable input/output pins on both sides of IC6.

Encoder (HT12E) and decoder (HT12D) ICs. The 2nd encoders are a series of CMOS LSIs for remote-control system applications. These are capable of encoding information that consists of N address bits and 12 N data bits. Each address/data input can be set to one of two logic states. Programmed addresses/data are transmitted together with header bits via an RF or infra-red transmission medium upon receipt of a trigger signal. The capability to select a TE trigger on HT12E or a data (DIN) trig-

ger on HT12D decoder further enhances the application flexibility of 2nd series of encoders. The HT12D also provides a 38kHz carrier for infra-red systems.

Transmitter. The transmitter consists of ATmega328 microcontroller (IC2), ADXL335 accelerometer, HT12E encoder (IC4) and 433MHz RF transmitter module (TX1). In this circuit, two analogue outputs from ADXL335 pins (x, y) are connected with input pins (23, 24) of the microcontroller. Analogue signals

are converted to digital signals through the microcontroller. Digital outputs from pins 16, 17, 18 and 19 of the microcontroller are directly sent to pins 13, 12, 11 and 10 of encoder IC4. This data is encoded and transmitted via RF module TX1.

Receiver. The receiver part consists of 433MHz RF receiver module (RX1), HT12D decoder (IC5) and L293D motor driver (IC6) to run the motors. Here, receiver module RX1 receives the transmitted signal, which is decoded by decoder IC to get the same digital outputs. Four outputs of IC6 drive two motors. The robot moves as per tilt direction of the accelerometer in the transmitter. The direction of the robot movement is as per logic listed in Table I.

Software program

The software program is written in Arduino programming language. We programmed a fresh ATmega328 microcontroller with the help of Arduino IDE 1.0.5 and an Arduino Uno board.

First, we have to load bootloader code into the microcontroller. For that, we used Arduino Uno for in-system programming (ISP) given in the IDE, by selecting File → Examples → Arduino ISP. Once the bootloader